

Project Description

The "Log Recon Analyzer" is a specialized forensic automation tool developed in Python. Its primary purpose is to parse, analyze, and extract actionable intelligence from Linux authentication logs (/var/log/auth.log). This tool was designed to assist cybersecurity teams in identifying security anomalies, tracking user behavior, and auditing system access in real-time.

Key Features Implemented:

1. **Privilege Escalation Audit:** The tool extracts and displays all sudo commands initiated by users, correctly identifying the source user versus the target user (Requirements 1.1-1.3, 2.5).
2. **Identity Lifecycle Tracking:** It monitors the creation and deletion of user accounts to detect unauthorized access or suspicious HR anomalies (Requirements 2.1-2.2).
3. **Authentication Monitoring:** Tracks sensitive events such as password changes and su (Switch User) sessions to ensure accountability (Requirements 2.3-2.4).
4. **Threat Detection & Alerting:** The tool identifies failed authentication attempts and correlates them with the attempted commands to detect potential brute-force attacks or unauthorized command execution (Requirement 2.6).

Technical Stack:

- **Language:** Python 3.x
- **Core Libraries:** re (Regular Expressions), argparse, os.
- **Logic:** The script utilizes advanced regex pattern matching to parse raw log lines and organizes the data into structured, professional ASCII tables.

Screenshot of Tool Execution:

kali-linuxOG - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- kali-linuxOG
- kaliNew
- Win10 victimAa1
- Python_Practice
- meta2

root@kali: /home/kali/scripts

Session Actions Edit View Help

LOG RECON ANALYZER | ENTERPRISE EDITION | v3.5.2
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— [PHASE 1: PRIVILEGE ESCALATION COMMANDS] —
Objective: Audit sudo command usage (Req 1.1-1.3, 2.5)

TIMESTAMP	SOURCE USER	COMMAND EXECUTED
Jan 31 09:00:01 2026	devops	/usr/bin/docker ps
Jan 31 09:15:22 2026	admin	/usr/bin/cat /etc/shadow
Jan 31 09:45:10 2026	service_acct	/usr/bin/systemctl reload nginx

— [PHASE 2: IDENTITY LIFECYCLE MANAGEMENT] —
Objective: Monitor user creation and deletion (Req 2.1, 2.2)

TIMESTAMP	EVENT TYPE	ACCOUNT NAME	DESCRIPTION
Jan 31 10:00:00 2026	USER_ADD	intern_user	New Account Created
Jan 31 10:05:00 2026	USER_ADD	contractor_tmp	New Account Created
Jan 31 11:30:00 2026	USER_DEL	contractor_tmp	Account Deleted

— [PHASE 3: AUTHENTICATION & SESSION EVENTS] —
Objective: Track password changes and SU switching (Req 2.3, 2.4)

TIMESTAMP	EVENT TYPE	SOURCE USER	DETAILS
Jan 31 10:10:00 2026	PASS_CHG	intern_user	Password Modified
Jan 31 10:20:00 2026	SU_SWITCH	devops	Switched to user: postgres

— [PHASE 4: SECURITY ANOMALIES & ALERTS] —
Objective: Detect failed authentication attempts (Req 2.6)

TIMESTAMP	LOGNAME (SOURCE)	ATTEMPTED COMMAND	STATUS
Jan 31 13:00:00 2026	unknown	/bin/bash	Failed Sudo Attempt
Jan 31 13:05:22 2026	backup_svc	/usr/bin/rsync -av /etc /tmp	Failed Sudo Attempt

— SUMMARY STATISTICS —
Total Events Processed: 10
Critical Alerts: 2

[+] Analysis Session Completed Successfully.

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.